

Design a sofa or chair around its user

Overall sofa dimensions differ from the effective seating dimensions experienced by a user. Link posture, support and cushion behaviour to a specification that can be tried and reviewed.

What you will accomplish

- Start with the user and task posture
- Distinguish overall and effective dimensions
- Record the trial and reference sample

Start with the user and task posture

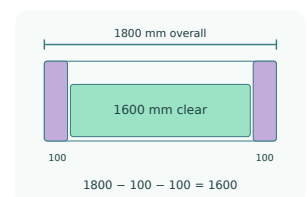
Define the task, sitting duration and how the user sits down and rises. In a mock-up, review foot and back support and the area behind the knee. Stature alone cannot establish a seat that suits everyone. Record observations and any access or further assessment needs rather than turning one user observation into a universal ideal dimension.



Task-specific user trials inform the seating specification.

Distinguish overall and effective dimensions

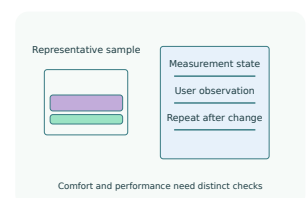
Overall width includes arms and overall depth includes the back. Draw a section showing the usable seat and back support with the cushion occupied. Geometry exercise only: an 1800 mm overall width with two 100 mm arms leaves 1600 mm clear. This arithmetic establishes neither comfortable occupancy nor effective loaded seat height.



Exercise: $1800 - 100 - 100 = 1600$ mm clear.

Record the trial and reference sample

Use a sample representing the proposed frame, support and filling, not an unrelated cushion on a random chair. Record effective dimensions, measurement state and sitting/rising observations. Changes to filling or back angle may change feel and usable dimensions, so repeat the trial after changes. Tie the specification to a reference sample and distinguish comfort acceptance from required strength and durability verification.



Comfort acceptance and structural performance are separate checks.

Worked case and decision

The 1800 mm exercise with two 100 mm arms gives 1600 mm clear. Enlarging each arm to 150 mm while retaining overall width reduces clear space to 1500 mm. Update the drawing and repeat the user trial before approving the new appearance: usable width has decreased by 100 mm despite an unchanged outside length.

Decision: record 1500 mm clear and repeat the seating trial.

Check understanding

1800 mm overall with two 150 mm arms: clear width?

- 1500 mm
- 1650 mm
- 1800 mm

Correct: 1500 mm. Subtract both arms: 1500 mm; this does not prove comfort.

Lesson questions

Do equal outside dimensions guarantee equal comfort?

No. Arms, back, filling and support change effective space and user experience.

Practice

Submit a user brief, seat section, width calculation and trial record separating comfort from strength verification.

User and task

Overall/clear dimensions and measurement state

Sample and trial results

Review your work against the criteria. Saving it does not constitute technical project approval.

- Both arms deducted
- Effective measurement state explicit
- No universal comfort claim